

Mapping Chlorophyll-a Concentrations in the Northwestern Iberian Peninsula

Noah Steixner
STXTAT001

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1 Introduction

Chlorophyll-a concentration in the ocean surface is an essential climate variable which provides crucial information regarding seasonal variations in phytoplankton abundance, which is closely linked to biogeochemical cycling more widely. Marine N and C cycles are of particular interest to climate science, as they allow for the estimation of carbon dioxide uptake, release and sequestration throughout seasonal cycles, as well as in response to changes in surface currents, wind forcing and nutrient input. Chlorophyll-a concentrations can be measured in-Situ or using remote sensing techniques. While in-Situ measurements are generally more accurate, they are difficult and resource-intensive to scale, and thus remote sensing techniques are generally preferred to generate global datasets (Sathyendranath et al., 2019). In this mapping exercise, an ESA CCI global ocean colour climatology spanning from September 1997 to August 1998 is used to map a section of the Northwestern Iberian Peninsula, in order to evaluate the scale and existence of seasonal variations, as well as to identify any overlap of chlorophyll-a concentrations with major bathymetric features.

The study site was selected due to its location along upwelling centers, and spans from Cape Finisterre in the North (43°N) to Porto in the South (41°N). The region is influenced by the Portugal current, a weak and seasonally variable current which is also considered the northward extent of the Canary current. The Canary current is the Eastern Boundary Current (EBC) that complements the poleward flow of the Gulf Stream in the west, and as is usual for EBCs, seasonal wind-driven upwelling is common. Along the Northwestern Iberian Peninsula, the upwelling season generally spans from June-September, when a northward shift of the Azores high leads to more frequent northerly winds in the region (Ferreira Cordeiro et al., 2018). The remainder of the year, from October to April/May, is generally considered the downwelling season (Otero et al., 2009). It was of interest to me whether these seasonal patterns are visible in the overall global dataset.

While the region spanning from Cape Sillero to Porto between 42°N and 41°N is well-known as an upwelling center, the embayments north of Cape Sillero are also an area where chlorophyll-a concentration may be seasonally elevated. This series of embayments, known as Rias Baixas, are an extensive area where water is shallow, and may be warmed rapidly in summer months (Otero et al., 2009). When this coincides with sufficient nutrient input, as may be the case with supply from inflowing rivers, conditions for algal growth are favourable. River plumes and sediment transported by major rivers such as the Minho, Douro, Vigo and Arousa rivers may lead to year-round elevated chlorophyll-a concentrations regionally, as rivers transport nutrients from terrestrial ecosystems upstream. High river discharge coupled with downwelling was shown to form a stratified layer of freshwater on the shelf, providing conditions conducive to phytoplankton growth (Otero et al., 2009).

Lastly, nutrient dynamics may be influenced by urban centres such as Porto or Vigo, and though wastewater and nutrient runoff from agricultural activities are reasonably well-managed, it is interesting to observe whether this urban imprint is visible in regional chlorophyll-a data.

2 Dataset Description

For the purpose of the mapping exercise, two datasets were used: the GMRT (Global Multi Resolution Topography) bathymetry dataset (Ryan et al., 2009) and the ESA CCI (Climate Change Initiative) ocean colour dataset (Sathyendranath et al., 2019), a synthesis product from a series of long-term satellite missions. These global datasets are publicly available and can be downloaded directly or through the use of an FTP request.

The GMRT bathymetry dataset collates data obtained from high-resolution multibeam echosounding surveys, which are only available for small portions of the seabed with lower resolution satellite-derived data as well as curated data contributed individually by researchers (Ryan et al., 2009). Satellite data uses anomalies in sea surface height to infer the presence of major bathymetric features (GEBCO Bathymetric Compilation Group, 2026). The GMRT synthesis allows for all data to be represented in a single gridded product, retaining high resolution where available, and downloadable in a range of formats compatible with common conventions. The dataset was downloaded using the Climate & Forecasting convention, importable as an xarray Dataset (Hoyer & Hamman, 2017). The dataset contains a single variable, altitude in meters, as defined on a global grid, at 90km intervals. Global attributes give information about version, history and projection used, while data variable attributes specify units (meters) and data range (-4516.72m to 1046.79m).

The ESA ICC ocean colour climatology is a 4km-gridded dataset which is a synthesis of remote sensing data from both European and North American satellite missions, using a range of different sensors. As the sensors operate at different bandwidths, a global synthesis needs to correct for inter-sensor biases, as well as for

atmospheric variability introduced by cloud cover (Sathyendranath et al., 2019). Data is cross-validated with in-Situ measurements and error estimates as well as bias is reported (Sathyendranath et al., 2019). The dataset is downloadable as a netcdf file, and contains a coordinate reference system as well as the chlorophyll-a data as variables, with a coordinate grid, and time dimension. The global attributes provide a description of version history, sources, license and sensor information as well as a summary which provides relevant details on procedures by which data from various sensors and datasets were merged. Data variable attributes provide information on units, grid mapping and ancillary variables. Chlorophyll-a concentrations are measured in mg/m^3 .

3 Results

The bathymetric map highlights the extent of the continental shelf, which extends 50-60km off the coast along Porto coastline (41°N), and narrows significantly to the north. The prominent Rias Baixas drop off to depths of 200m at a similar distance from the coast, yet show an extended area of depths below 50m where embayments restrict current flow. It is of note that the continental shelf generally drops off steeply from depths of 200m to a depth of 1000m, which further to depths of 2000m along the continental rise. The Porto canyon is visible as a very deep region reaching up to 4000m of depth at around 41°N, 10.5°W. River sediment plumes are not visible.

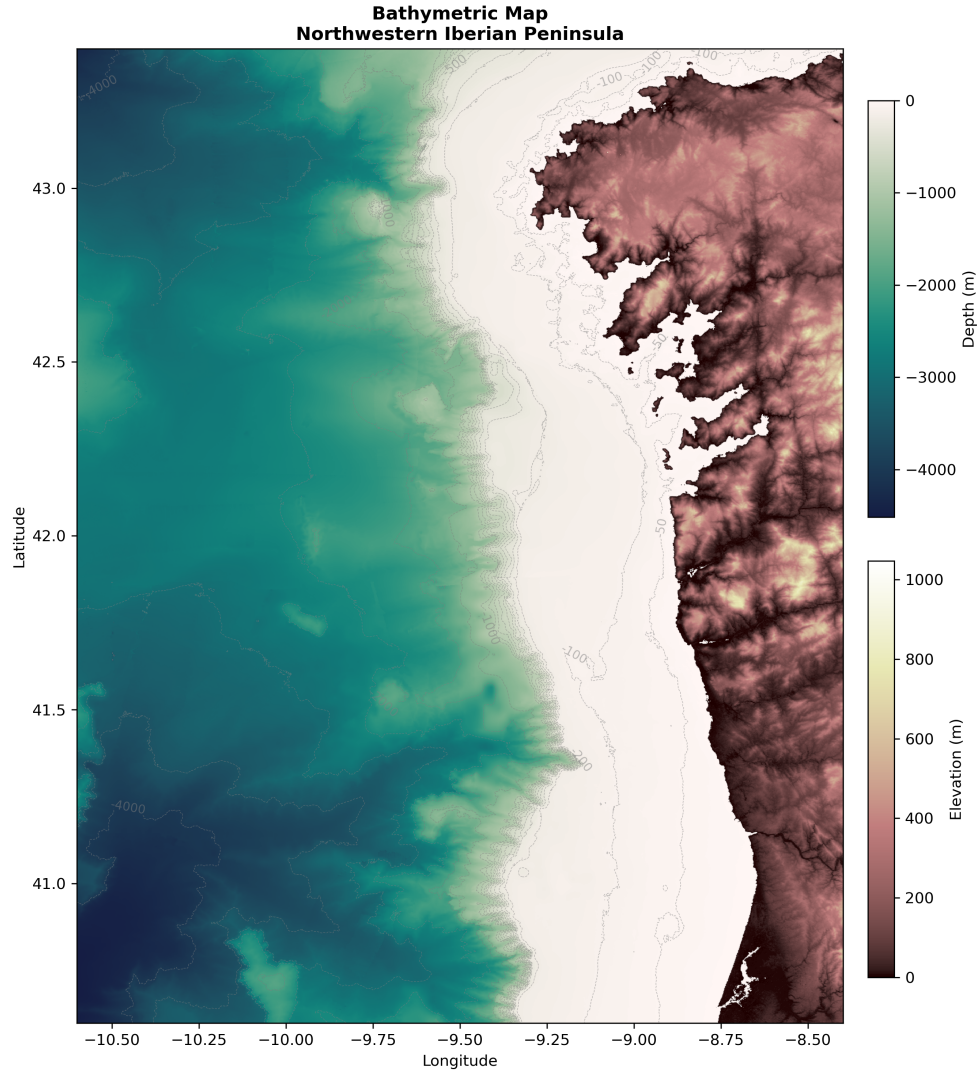


Figure 1: Bathymetric Map of the Northwestern Iberian Peninsula.
Section Extent: 43.4°N, 40.6°N, 10.6°W, 8.4°W.

Generally speaking, shallower depths are more readily warmed in summer months, and the rapid transition of the continental shelf into the depths promotes upwelling, as deeper water can replace the shallow waters of the continental shelf. These are two contrasting effects that may contribute to elevated chlorophyll-a concentrations in summer months.

Yearly averaged chlorophyll-a data for the region indicate that chlorophyll-a concentrations are generally higher in coastal areas, with particularly high values in the proximity of major river outlets. The Minho and Douro estuaries are the areas of highest chlorophyll-a concentrations, with intermediate concentrations observed along the Rias Baixas.

Chlorophyll-a values range from 0.30 – 8.88 mg/m^3 , with a median of 0.45 mg/m^3 . Thus, higher chlorophyll-a concentrations occur only in few grid cells in proximity to the coastline. Chlorophyll-a concentrations over the open ocean are generally low, as is expected in this region.

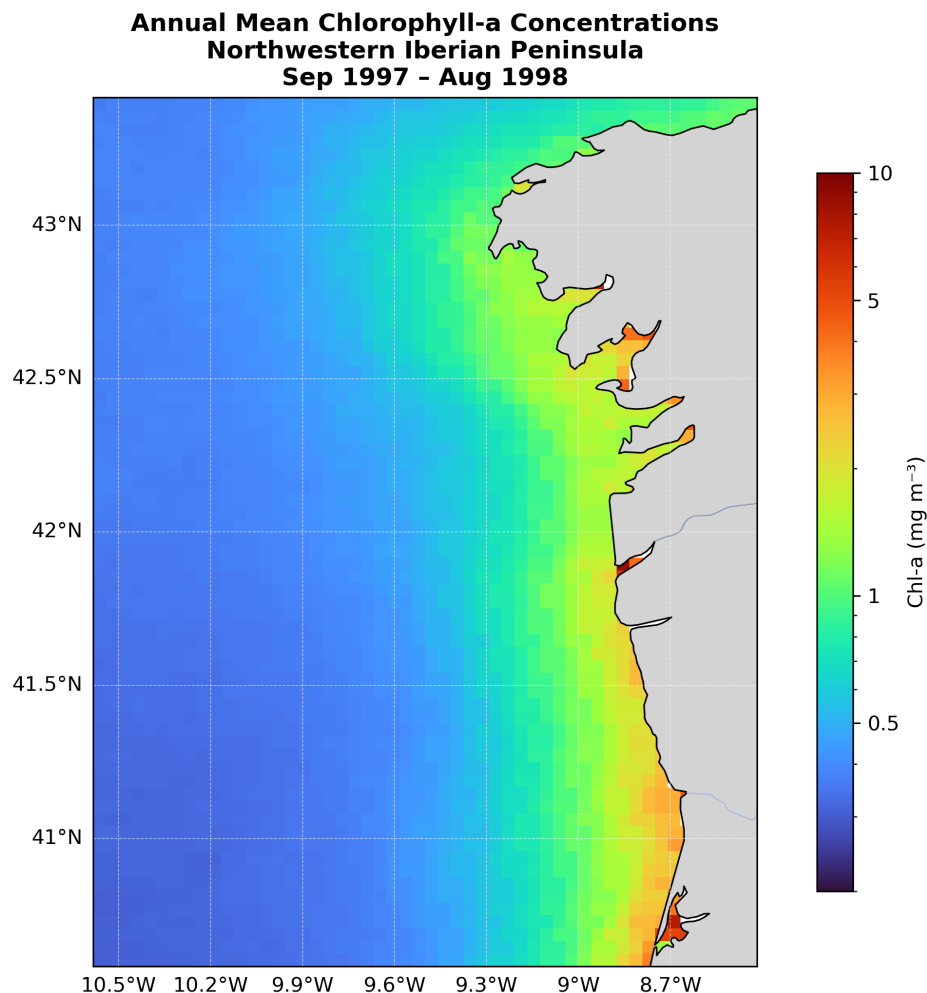


Figure 2: Annual Mean of Chlorophyll-a Concentrations
September 1997 to August 1998.

Monthly plots show that contrary to an expected maximum during summer months, when upwelling is common, chlorophyll-a concentrations over the open ocean peak during spring in this dataset. This may be due to nutrient mobilisation and the onset stratification after winter mixing. However, it is surprising that chlorophyll-a concentrations remain elevated throughout winter months and tend to decrease during the summer (note: plotting order was checked). While the Rias Baixas display above-average chlorophyll-a concentrations for most of the year, the coastal upwelling region in the proximity of Porto (possibly in combination with its urban imprint) appear to be the chlorophyll-a hotspot in this dataset.

Monthly Mean Chlorophyll-a Concentrations Northwest Iberian Peninsula

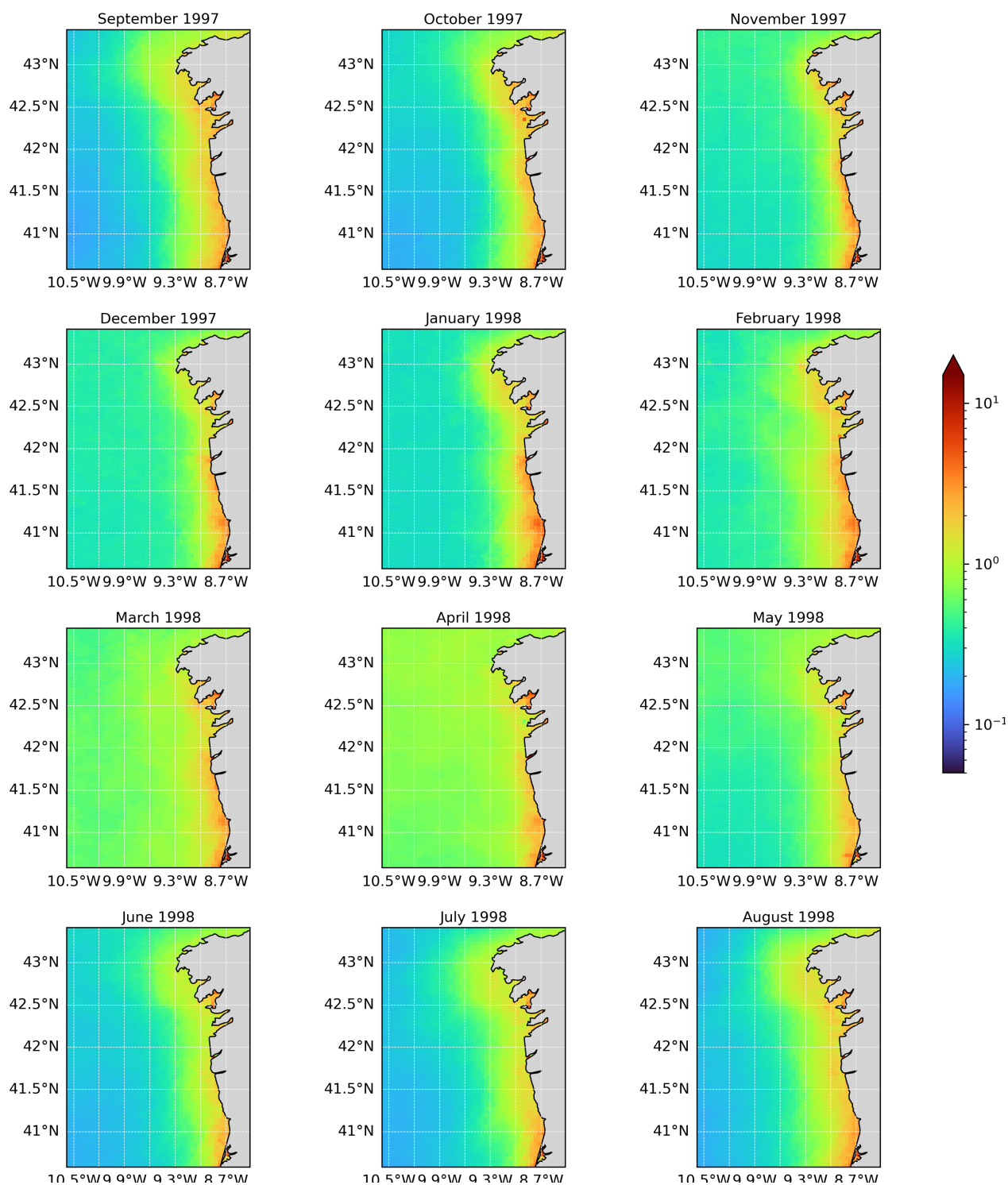


Figure 3: Monthly Mean of Chlorophyll-a Concentrations.
Chlorophyll-a concentrations are elevated during a spring phytoplankton bloom.

Lastly, a plot of the overall regional seasonal mean indicates that on average, chlorophyll-a concentrations generally peak in March/April, and reach a minimum in June, which is consistent with the monthly plots. In order to complement the regional average, a point that lies in close proximity to the Minho river (annual discharge: $\sim 9.5 \times 10^9 \text{ m}^3$) (Sousa et al., 2014) was selected for comparison. It is of note that at this point, chlorophyll-a concentrations appear to peak earlier, possibly due to elevated freshwater input from the Minho River, as indicated in Otero et al. (2009).

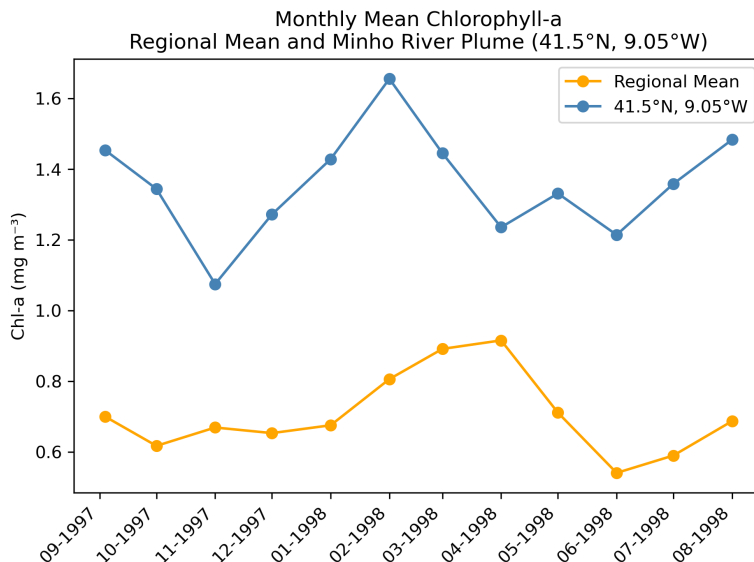


Figure 4: Monthly Mean Chlorophyll-a Concentrations.

Plot showing monthly means over the whole study area and a point influenced by the inflow of the Minho River into the Atlantic.

4 AI Credit

Besides following tutorials as provided in the SAMOS Core 3 Python in Ocean Sciences module (available at <https://github.com/mvichi/SAMOS-Core3-UCT>), Anthropic Claude using Sonnet 4.6 was used to provide outlines for bathymetry and annual mean plotting including separation of ocean and terrestrial variables, as well as for the axes separation approach in the faceted monthly mean plot. Further, AI was used to assist in colourbar scaling and integration into plots, and to validate steps in data analysis (Anthropic, 2025). Key prompts are available with submission.

Python modules used include xarray (Hoyer & Hamman, 2017), matplotlib (Hunter, 2007), and cartopy (Met Office, 2010). Official matplotlib docs were used to validate plotting approaches, e.g. (Matplotlib Development Team, 2025).

AI was not used to draft or improve the report, and basic approaches to data analysis were as introduced throughout lectures.

References

- Anthropic. (2025). *Claude Sonnet 4.5* [Large language model used for assistance with Python-based geospatial data analysis, including: logarithmic-scale Chlorophyll-a mapping with Matplotlib and Cartopy; reconstruction of malformed GMRT bathymetry grids from NetCDF4 files using xarray; adaptation of Cartopy projections to xarray faceted plots; and construction of time-series plots of monthly mean Chlorophyll-a concentrations]. <https://www.anthropic.com>
- Ferreira Cordeiro, N. G., Dubert, J., Nolasco, R., & Desmond Barton, E. (2018). Transient response of the north-western iberian upwelling regime. *PLoS One*, 13(5), e0197627.
- GEBCO Bathymetric Compilation Group. (2026). *The gebco 2026 grid – a continuous terrain model for oceans and land at 15 arc-second intervals*. NERC EDS British Oceanographic Data Centre, NOC. <https://doi.org/10.5285/4f68d5c7-45eb-f999-e063-7086abc036fa>
- Hoyer, S., & Hamman, J. (2017). Xarray: N-D labeled arrays and datasets in Python. *Journal of Open Research Software*, 5(1), 10. <https://doi.org/10.5334/jors.148>
- Hunter, J. D. (2007). Matplotlib: A 2d graphics environment. *Computing in Science & Engineering*, 9(3), 90–95. <https://doi.org/10.1109/MCSE.2007.55>
- Matplotlib Development Team. (2025). Matplotlib.pyplot.pcolormesh — matplotlib documentation [Accessed: 2026-05-13]. https://matplotlib.org/stable/api/_as_gen/matplotlib.pyplot.pcolormesh.html
- Met Office. (2010). Cartopy: A cartographic python library with a matplotlib interface [Accessed: 2026-05-13]. <https://scitools.org.uk/cartopy>
- Otero, P., Ruiz-Villarreal, M., & Peliz, Á. (2009). River plume fronts off nw iberia from satellite observations and model data. *ICES Journal of Marine Science*, 66, 1853–1864. <https://doi.org/10.1093/icesjms/fsp156>
- Ryan, W., Carbotte, S., Coplan, J., O'Hara, S., Melkonian, A., Arko, R., Weissel, R., Ferrini, V., Goodwillie, A., Nitsche, F., Bonczkowski, J., & Zemsky, R. (2009). Global multi-resolution topography synthesis. *Geochemistry Geophysics Geosystems*, 10. <https://doi.org/10.1029/2008GC002332>
- Sathyendranath, S., Brewin, R. J., Brockmann, C., Brotas, V., Calton, B., Chuprin, A., Cipollini, P., Couto, A. B., Dingle, J., Doerffer, R., Donlon, C., Dowell, M., Farman, A., Grant, M., Groom, S., Horseman, A., Jackson, T., Krasemann, H., Lavender, S., ... Platt, T. (2019). An ocean-colour time series for use in climate studies: The experience of the ocean-colour climate change initiative (oc-cci). *Sensors*, 19(19). <https://doi.org/10.3390/s19194285>
- Sousa, M., Vaz, N., Alvarez, I., Gomez-Gesteira, M., & Dias, J. (2014). Modeling the minho river plume intrusion into the rias baixas (nw iberian peninsula). *Continental Shelf Research*, 85, 30–41. <https://doi.org/https://doi.org/10.1016/j.csr.2014.06.004>